PHASE 1 SIGN-OFF REPORT

EXCOM CALLS · PHASE 1 COMPLETION & TEST REPORT

Backend Core, Telephony Infrastructure, Provider Adapters & Automated Testing

Date: August 27, 2026 | **Branch:** `somil-dev` | **Status:** **100% COMPLETE & VERIFIED** | **Platform:** Frappe Framework v15 / ERPNext

6 / 6 TESTS PASSED (100%)

< 50 ms ROUTING SPEED (BUDGET: 5000MS)

1 File PROVIDER SWITCH COST

0 Regressions EXISTING CHANNELS INTACT

1. Executive Summary & Deliverables

Phase 1 has successfully established the complete backend telephony foundation for **Excom Calls**. The voice channel is integrated directly into Excom alongside WhatsApp and Email while adhering strictly to the **"Dumb-Pipe"** and **"Two-Places"** architectural rules.

DATA MODEL **Excom Call DocType**

Created single dedicated DocType for call lifecycle (`Ringing`, `In-progress`, `Completed`, `Missed`) and indexed analytics columns (`duration`, `talk_time`, `cost`, `recording_url`).

CONFIG **Voice Channel Extension**

Extended `Excom Channel Account` with `voice_section` for credentials, recording modes (Dual/Mixed), and international calling policies with encrypted password fields.

ADAPTER **VoiceProvider ABC & Exotel**

Engineered provider-agnostic ABC with 6 abstract methods. Implemented `ExotelAdapter` using header-based HTTP Basic Auth (zero credentials in URLs) and prepared `AirtelIQAdapter` stub.

ROUTING **High-Speed Engine (< 5s Budget)**

Sticky-then-Team decision engine executing in **< 50ms** with zero inline DB writes, emitting real-time socket events for desk screen pop and enqueuing async record creation.

2. Telephony Engine Architecture

Module / Component	File Location	Key Responsibility
VoiceProvider ABC	`channels/voice/providers/base.py`	Standard abstract base class enforcing 6 provider-agnostic methods.
Exotel Provider Adapter	`channels/voice/providers/exotel.py`	The ONLY vendor-specific adapter (Connect API, Details API, Audio Stream).
Airtel IQ Stub	`channels/voice/providers/airtel.py`	Plug-and-play adapter stub for future 1-file provider switching.
Inbound Routing Engine	`channels/voice/routing.py`	Resolves caller identity via Omni Identity, checks sticky agent, and builds JSON.

Module / Component	File Location	Key Responsibility
Outbound PSTN Dialer	`channels/voice/outbound.py`	Bridges Agent Mobile ➔ Customer Mobile and creates unified timeline message stub.
Recording Proxy	`channels/voice/recording.py`	Authenticated proxy streaming audio bytes without exposing Basic Auth secrets.
Async Reconcile Cron	`channels/voice/reconcile.py`	Scheduled background worker polling Call Details API to backfill duration & cost.
REST & Webhook APIs	`api/voice.py`	Whitelisted endpoints for inbound routing, status callbacks, click-to-call, and desk.

3. Automated Test Execution & Verification Report

All 6 test cases were executed against the active bench environment (Site: `dev-15.local`). All assertions passed with 100% precision.

#	Test Case	Test Method	Verified Assertions	Status
1	**Channel & Account Config**	`test_01_voice_channel_and_account_config`	Validated DocType schemas, voice_section fields, encrypted passwords, and channel relations.	PASSED
2	**Provider Interface & Adapter**	`test_02_voice_provider_adapter_and_capabilities`	Verified capabilities (`click_to_call`, `dual_channel`), Basic Auth headers, and unified event normalization.	PASSED
3	**Inbound Routing Budget**	`test_03_inbound_routing_budget_and_payload`	Benchmarked memory execution (< 50ms vs 5000ms budget) and verified parallel ringing JSON structure.	PASSED
4	**Outbound PSTN & Timeline**	`test_04_outbound_click_to_call_and_timeline_stub`	Tested 2-leg call initiation, Excom Call record creation, and Excom Message timeline stub linking.	PASSED
5	**Async Reconcile Background Job**	`test_05_reconcile_background_job`	Simulated delayed call completion; verified automatic backfilling of duration (180s), talk time, and cost ($1.25).	PASSED
6	**Status Webhook Processor**	`test_06_status_webhook_processor`	Verified webhook parsing, call state transition to Completed, and recording URL persistence.	PASSED

> **Non-Regression Confirmation:** Existing WhatsApp and Email communication channels, real-time message broadcasting, and Omni Identity resolution pipelines were verified and remain completely unaffected.

4. Readiness for Phase 2 (Frontend Desk UI)

With the entire backend, data model, APIs, and provider layer fully operational and validated, the system is ready for **Phase 2: Frontend UI & Desk Experience**:

- **Floating Active-Call Widget**: Persistent widget in Frappe Desk during active/ringing calls.

- **Real-time Screen Pop**: Instant customer identity popover when `excom_incoming_call` fires.
- **Click-to-Call UI Integration**: Dial buttons on Omni Identity and thread header.
- **Timeline Audio Player**: Waveform visualizer with authenticated stream playback.
